

Giulia Rocco

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ABOUT

Biomedical Engineer and MSCA Fellow specializing in multimodal neuroimaging (fNIRS, fMRI, EEG) and advanced signal processing. Expertise in brain functional mapping and the development of tailored analytical pipelines for high-dimensional neural data. Proven track record in leading international collaborations (Canada, France, Italy) and securing competitive funding. Current research focuses on calibrated fMRI and optical imaging to quantify cerebrovascular function and oxygen metabolism in healthy and clinical populations.

EXPERIENCE

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| Institute for Advanced Biomedical Technologies , Postdoctoral researcher | Chieti, Italy
Mar 2025 – present |
| <ul style="list-style-type: none">◦ Multimodal data collection and analysis (fMRI, fNIRS) with healthy controls and multiple sclerosis patients◦ Coordinated cross-institutional workflows, streamlining scientific and administrative milestones to ensure project deadlines were met | |
| I3S Lab and Centre Hospitalier Universitaire de Nice , Graduate Research Assistant | Nice, France
Jan 2020 – Nov 2023 |
| <ul style="list-style-type: none">◦ Conducted multimodal data collection and analysis (fMRI, fNIRS, EEG) to investigate cerebellar activation, resulting in 6 published findings◦ Designed and validated non-invasive novel experimental frameworks for studying cerebellum, earning awards for breakthroughs in cerebellar neuroscience◦ Led cross-institutional project management, coordinating 50+ subject recruitment, 20k\$ budget oversight, and data integrity, while also successfully obtaining two equipment loans from industry to facilitate experiments | |
| NIRx Medical Technologies , PhD Intern | Berlin, Germany
June 2022 – July 2022 |
| <ul style="list-style-type: none">◦ Developed a customer-facing document detailing the value proposition of simultaneous fNIRS/fMRI, including hardware and software summaries, and drafted a video plan for data analysis software promotion | |
| Concordia University, MultifunkIM Lab , Visiting Researcher | Montreal, Canada
Mar 2022 – Dec 2022 |
| <ul style="list-style-type: none">◦ Pioneered a computational framework for cerebellar fNIRS image reconstruction, effectively solving forward and inverse problems◦ Led multimodal data collection projects involving 30 participants and 300+ recordings | |
| Université Côte d'Azur , Teaching Assistant | Nice, France
Sept 2020 – Dec 2020 |
| <ul style="list-style-type: none">◦ Taught Algorithms & Programming with R to 30+ students, delivering practical sessions and project-based assignments | |
| Politecnico di Milano, B3Lab and PHEEL Laboratories , Research Intern | Milan, Italy
Jan 2019 – Nov 2019 |
| <ul style="list-style-type: none">◦ Analyzed biomedical signals (EEG, ECG, PPG, EDA, eye tracking) for emotion and stress detection, developing novel indices for biomedical and biomarketing applications◦ Contributed to multidisciplinary VR therapy research for neurodevelopmental disorders, managing physiological monitoring and data analysis◦ Conducted a comparative analysis of two popular wearable devices and a clinical-grade ECG, revealing significant variations in inter-beat-interval accuracy | |
| KU Leuven, Audio Engineering Lab , Visiting Scholar | Leuven, Belgium
Oct 2017 – Mar 2018 |
| <ul style="list-style-type: none">◦ Design and implementation of a novel language-independent speech intelligibility test for hearing screening, resulting in 3 publications◦ Characterized speech stimuli intelligibility and psychometric functions with different noise realizations | |
| Lutech , Support Engineer | Milan, Italy
Oct 2015 – Dec 2015 |
| <ul style="list-style-type: none">◦ Support to staff in operating rooms within the development of an EHR framework | |
| Università di Milano, LAFAS Lab , Visiting Student | Milan, Italy
Mar 2015 – July 2015 |
| <ul style="list-style-type: none">◦ Validated and optimized machine learning algorithms for CT image segmentation, achieving a correlation of over 0.9 with manual methods and reducing processing time by 53% | |

EDUCATION

- Université Côte d'Azur, Nice, France** — PhD in Automatics, Signal and Image processing Jan 2020 – Nov 2023
- Marie Curie fellow - **BoostUrCAreer** program
 - Thesis: *"Multimodal Functional Exploration of the Cerebellum"*
- Politecnico di Milano, Milan, Italy** — MSc in Biomedical Engineering Sept 2015 – Apr 2018
- Final Grade: 110/110
 - Thesis Research at KU Leuven: *"Design, implementation, and pilot testing of a language-independent speech intelligibility test"*
 - Erasmus+ Exchange at KTH Royal Institute of Technology, Stockholm.
- Politecnico di Milano, Milan, Italy** — BSc in Biomedical Engineering Sept 2012 – July 2015
- Final Grade: 102/110
 - Thesis: *"Validation and optimisation of an automatic hard tissue segmentation algorithm for CBCT data"*

SKILLS

Programming: Matlab, R, C, Python
Data analysis: SPM, FreeSurfer, FSL, Brainstorm, Fieldtrip, Homer3, EEGLAB
Languages: Italian (native), English, French

FUNDING

2023: Horizon 2020 MSCA COFUND doctoral scholarship extension (3 months)
2022: Quebec Bio-Imaging Network grant for Pilot Projects (\$20,000) - Conceived the project and drafted the application
2020: Horizon 2020 MSCA COFUND doctoral scholarship (42 months) - 15 students selected over 500 applicants
2019: Biomedical Engineering departmental funds for research interns, Politecnico di Milano
Merit scholarship for master thesis abroad, Politecnico di Milano
2017: Erasmus+ mobility scholarship (2 spots for KTH for Biomedical Engineering), Politecnico di Milano
2015: Merit scholarship for off-campus students, Politecnico di Milano

AWARDS

2023: Nomination for the best PhD thesis prize of the doctoral school
2021: Public Award, 3-minutes Thesis contest in 18th IEEE International Symposium on Biomedical Imaging
Best Poster Award, 4th Congress of the Physiology and Integrative Biology French Society
Winner, MSCA Falling Walls Lab 2021: Breaking the wall of Cerebellar Neuroscience (Top 75 Emerging Talents worldwide)
Université Côte D'Azur Excellence Prize
2012: Winner, Literary Contest - An Author to Rediscover – Sibilla Aleramo: A Journey Called Woman
Winner, Latin Contest - Certamen Leonianum
Honorable Mention, Latin Contest - Certamen Horatianum

TEACHING

2025: Teaching Assistant, Industrial Bioengineering & Bioengineering and Biomechanics, University of Chieti-Pescara
Teaching Assistant, Physics, University of Chieti-Pescara
2024: Co-instructor, "NIRSTORM: a Brainstorm plugin for fNIRS statistical analysis", Educational course at SfNIRS 2024
2022: Co-instructor, "NIRSTORM: a Brainstorm plugin for fNIRS statistical analysis", Educational course at SfNIRS 2022
2020: Teaching Assistant, Algorithms & Programming with R, Université Côte D'Azur

TALKS AND PRESENTATIONS

- 2026: Poster**, International Society for Magnetic Resonance in Medicine (ISMRM)
Simultaneous gas-free calibrated fMRI and fNIRS measurements in human visual cortex: a feasibility study
- 2025: Poster**, fNIRS UK Biennial meeting
Simultaneous fNIRS, BOLD, and CBF Measurements on the Visual Cortex: a feasibility study
Poster, fNIRS Italy Biennial meeting
Toward simultaneous calibrated fMRI and fNIRS acquisitions: a case study in the visual cortex
- 2024: Oral talk**, Biennial meeting of the Society of fNIRS (SfNIRS)
Enhancing Cerebellar fNIRS/fMRI via Tailored Pipelines
Invited talk, RWTH-NIRx fNIRS workshop
Simultaneous fNIRS/fMRI exploration of the cerebellum
- 2022: Poster**, Biennial meeting of the Society of fNIRS (SfNIRS)
Densifying Optodes Montage to Enhance Cerebellar fNIRS
Poster, 19th IEEE International Symposium on Biomedical Imaging (ISBI)
When fNIRS meets fMRI to complement cerebellar exploration
- 2021: Poster**, 4th Congress of the Physiology and Integrative Biology French Society
Asymmetries in Cerebellar Activation during Finger Movements: A Functional Near-Infrared Spectroscopy Study
Paper presentation, 43rd IEEE Engineering in Medicine and Biology Society (EMBC)
A chiral fNIRS spotlight on cerebellar activation in a finger tapping task
Poster, 18th IEEE International Symposium on Biomedical Imaging (ISBI)
Exploring the cerebellum with functional near-infrared imaging: a preliminary study
- 2020: Paper presentation**, 42nd IEEE EMBC
Exploration of the physiological response to an online gambling task by frequency domain analysis of the electrodermal activity
Poster, SophIA Summit
P300 event-related potentials classification from EEG data through interval feature extraction and recurrent neural networks

CONFERENCE ORGANIZATIONS AND RESPONSIBILITIES

- 2027:** fNIRS Italy Biennial meeting, Pescara - Organising Committee
- 2026:** Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS), Macau - Conference support team
- 2022:** Jury member for MSCA Falling Walls Lab
- 2021:** BoostUrCareer Doctoriales - Program Committee

PROFESSIONAL FORMATION

- 2023:** Introduction to ethics and scientific integrity
- 2022:** PhD Career development, Association Bernard Gregory (ABG)
- 2021:** TED-worthy presenting, Debatrix - funded by MSCA Falling Walls Lab
BootCamp Open Science, OPTIMA, TU Graz - Best practices

OUTREACH

- 2025:** European Researcher Night, University of Chieti-Pescara - The physics behind MRI
Tales from MEDNIGHT - Translator of short science-inspired tales for kids
- 2024:** Tales from MEDNIGHT - Translator of short science-inspired tales for kids
- 2023:** Brain Week, Université Côte D'Azur - Introduction to brain measurements
Secondary School Outreach Mentor, Université Côte D'Azur - Practical laboratory on optical imaging
- 2022:** International Day of Women and Girls in Science - Interviewed by MSCA, European Commission
Secondary School Outreach, Université Côte D'Azur - Introduction to brain function and optical imaging
- 2021:** European Researcher Night, Université Côte D'Azur - Researcher's suitcase
- 2020:** European Researcher Night, Université Côte D'Azur - Science comics

PUBLICATIONS

Simultaneous fNIRS-fMRI in Cerebellum and Motor Cortex Reveals High Temporal Correlation with Pipeline-Dependent Spatial Variability
Rocco, G., Delaire, E., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J.

In preparation

Surface-based Analysis of Cerebellar Personalized High-Density fNIRS and fMRI During a Motor and an Auditory Task <u>Rocco, G.</u> , Delaire, E., Cai, Z., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J.	<i>In preparation</i>
Quantitative functional BOLD (qfBOLD): A Gradient-Echo/Spin-Echo Framework for Oxygen Extraction Fraction (OEF) Mapping with functional MRI Chiarelli, A. M., Chalet, L., Pomante, S., Di Censo, D., Caporale, A., Biondetti, E., Fasano, F., Zaca', D., <u>Rocco, G.</u> , Carriero, M., Graziano, F., Fear, E. J., Caligiuri, M. E., Wise, R., Germuska, M. (Journal of Cerebral Blood Flow and Metabolism)	<i>Under review</i>
Simultaneous gas-free calibrated fMRI and fNIRS measurements in human visual cortex: a feasibility study <u>Rocco, G.</u> , Chalet, L., Pomante, S., Fear, E. J., Graziano, F., Di Censo, D., Carriero, M., Perpetuini, D., Del Gratta, C., Perrucci, M. G., Wise, R., Chiarelli, A. M. (International Society for Magnetic Resonance in Medicine (ISMRM))	May 2026
Assessing the Minimum Number of Breath-Holds Required for Stable CMRO₂ and OEF Estimates Using Breath-Hold Calibrated fMRI Fear, E. J., Di Censo, D., Pomante, S., Carriero, M., Graziano, F., Chalet, L., <u>Rocco, G.</u> , Caporale, A., Biondetti, E., Fasano, F., Zaca', D., Tomassini, V., Chiarelli, A. M., Wise, R. (International Society for Magnetic Resonance in Medicine (ISMRM))	May 2026
Mapping Cerebrovascular Reactivity: Comparison of Gradient Echo and Spin Echo BOLD with Arterial Spin Labelling Pomante, S., Di Censo, D., Caporale, A., Fasano, F., Zaca', D., Fear, E. J., Graziano, F., Carriero, M., <u>Rocco, G.</u> , Chalet, L., Biondetti, E., Caligiuri, M. E., Germuska, M., Wise, R., Chiarelli, A. M. (International Society for Magnetic Resonance in Medicine (ISMRM))	May 2026
Assessing Cerebral Oxygen Metabolism in Multiple Sclerosis Using Breath-Hold single Calibrated fMRI Di Censo, D., Fear, E. J., Caporale, A., Censi, S., Graziano, F., Biondetti, E., Chalet, L., <u>Rocco, G.</u> , Carriero, M., Pomante, S., Patitucci, E., Fasano, F., Zaca', D., Germuska, M., Chiarelli, A. M., Tomassini, V., Wise, R. (International Society for Magnetic Resonance in Medicine (ISMRM))	May 2026
Quantitative Functional BOLD (qfBOLD) for Oxygen Extraction Fraction Mapping Using ASL-free Gradient Echo/Spin Echo fMRI Chiarelli, A. M., Chalet, L., Pomante, S., Di Censo, D., Caporale, A., Biondetti, E., Fasano, F., Zaca', D., <u>Rocco, G.</u> , Carriero, M., Graziano, F., Fear, E. J., Caligiuri, M. E., Wise, R., Germuska, M. (International Society for Magnetic Resonance in Medicine (ISMRM))	May 2026
Integrating a flow-diffusion model of O₂ transport to qBOLD for contrast-agent- and gas-free mapping of deoxyCBV and CMRO₂ Chalet, L., Di Censo, D., Pomante, S., <u>Rocco, G.</u> , Fear, E. J., Graziano, F., Carriero, M., Fasano, F., Biondetti, E., Perpetuini, D., Wise, R., Chiarelli, A. M. (International Society for Magnetic Resonance in Medicine (ISMRM))	May 2026
Long Short-Term Memory Networks for Fast Optical Signal Identification in the Human Visual Cortex for Brain Computer Interface Applications Perpetuini, D., <u>Rocco, G.</u> , Fear, E., Pomante, S., Chalet, L., Graziano, F., Valeri, E., Carriero, M., Del Gratta, C., Perrucci, M. G., Wise, R., Chiarelli, A. M. (International Conference on e-Health and Bioengineering (EHB))	Nov 2025
Enhancing Cerebellar fNIRS/fMRI via Tailored Pipelines <u>Rocco, G.</u> , Delaire, E., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J. (Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS))	Sept 2024

Characterization of the Intelligibility of Vowel–Consonant–Vowel (VCV) Recordings in Five Languages for Application in Speech-in-Noise Screening in Multilingual Settings <u>Rocco, G.</u> , Bernardi, G., Ali, R., van Waterschoot, T., Polo, E. M., Barbieri, R., Paglialonga, A. (Applied Sciences)	May 2023
Densifying Optodes Montage to Enhance Cerebellar fNIRS <u>Rocco, G.</u> , Delaire, E., Ramanoël, S., Meste, O., Magnié-Mauro, M. N., Grova, C., Lebrun, J. (Biennial meeting of the Society of Functional Near-Infrared Spectroscopy (SfNIRS))	Oct 2022
When fNIRS meets fMRI to complement cerebellar exploration <u>Rocco, G.</u> , Ramanoël, S., Habas, C. N., Arleo, A., Meste, O., Magnié-Mauro, M. N., Lebrun, J. (IEEE International Society for Biomedical Imaging (ISBI))	Mar 2022
A chiral fNIRS spotlight on cerebellar activation in a finger tapping task <u>Rocco, G.</u> , Lebrun, J., Meste, O., Magnié-Mauro, M. N. (43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))	Nov 2021
Single-Trial Detection of Event-Related Potentials with Integral Shape Averaging: An Application to the Elusive N400 <u>Rocco, G.</u> , Rix, H., Lebrun, J., Guetat, S., Chanquoy, L., Meste, O., Magnié-Mauro, M. N. (43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))	Nov 2021
Asymmetries in Cerebellar Activation during Finger Movements: A Functional Near-Infrared Spectroscopy Study <u>Rocco, G.</u> , Lebrun, J., Meste, O., Magnié-Mauro, M. N. (Acta Physiologica: 4th Congress of Physiology and Integrative Biology)	Sept 2021
Exploring the cerebellum with functional near-infrared imaging: a preliminary study <u>Rocco, G.</u> , Lebrun, J., Meste, O., Magnié-Mauro, M. N. (IEEE 18th International Society for Biomedical Imaging (ISBI))	Apr 2021
P300 event-related potentials classification from EEG data through interval feature extraction and recurrent neural networks <u>Rocco, G.</u> , Lebrun, J., Magnié-Mauro, M. N., Meste, O. (SophIA Summit)	Nov 2020
An automated speech-in-noise test for remote testing: Development and preliminary evaluation Paglialonga, A., Polo, E. M., Zanet, M., <u>Rocco, G.</u> , van Waterschoot, T., Barbieri, R. (American Journal of Audiology)	Sept 2020
Exploration of the physiological response to an online gambling task by frequency domain analysis of the electrodermal activity <u>Rocco, G.</u> , Reali, P., Lolatto, R., Tacchino, G., Mandolfo, M., Mazzola, A., Bianchi, A. M. (42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))	July 2020
Characterization of eye gaze and pupil diameter measurements from remote and mobile eye-tracking devices Lolatto, R., <u>Rocco, G.</u> , Mustoni, R., Maninetti, C., Pastura, R., Pigazzini, A., Barbieri, R. (XV Mediterranean Conference on Medical and Biological Engineering and Computing (MEDICON))	Jan 2020
Development and preliminary evaluation of a novel adaptive staircase procedure for automated speech-in-noise testing Zanet, M., Polo, E. M., <u>Rocco, G.</u> , Paglialonga, A., Barbieri, R. (41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC))	July 2019

Heart rate variability from wearables: a comparative analysis among standard ECG, a smart shirt and a wristband

June 2019

Realì, P., Tacchino, G., Rocco, G., Cerutti, S., Bianchi, A. M.
(pHealth 2019)

Validation of an automatic hard tissue segmentation algorithm for cone beam CT data

Aug 2015

Codari, M., Caffini, M., Rizzo, L., Rocco, G., Tartaglia, G. M., Baselli, G., Sforza, C.
(37th IEEE Annual conference on Engineering in Medicine and Biology)